

Auto Ticket Classification using Flow Designer

Problem Statement

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems, and slow computers.

Currently, IT staff manually reviews each request and assigns a category, which is time-consuming and inefficient.

This project automates ticket classification using ServiceNow Flow Designer by analyzing keywords in the incident's short description and description fields.

Objective

- Automatically classify incidents at the time of creation.

- Reduce manual effort for IT agents.

- Improve ticket routing efficiency.

- Implement a no-code, easily maintainable solution.

PHASE 1: Requirement Analysis & Planning

The school IT helpdesk receives multiple support requests daily from students and teachers related to common issues such as network connectivity, hardware failures, account access problems, and system performance issues.

Currently, IT staff manually review each ticket to:

- Identify the issue type.

- Select the correct category and subcategory.

- Inform the caller about ticket creation.

This manual process is time-consuming, error-prone, and not scalable as the number of requests increases.

Activity : 1 Business Use Case.

Business Requirements

The system must:

1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format.
6. Ensure easy maintenance and future scalability.

Activity : 2 Creation of New Update Set.

Navigation Steps :

Go to All menus → Search for Update set.

Under System Update Sets → Select Local Update Sets.

Click the New button on the top right corner of the page.

Name : **Project Update Set.**

State : **In progress.**

Application : **Global.**

Then **Save** the **Form.**

Under Related links Click on the **Make This My Current Set .**

Then the Created Update Set will appear as Current.

Activity : 3 Navigation steps in Picture View.

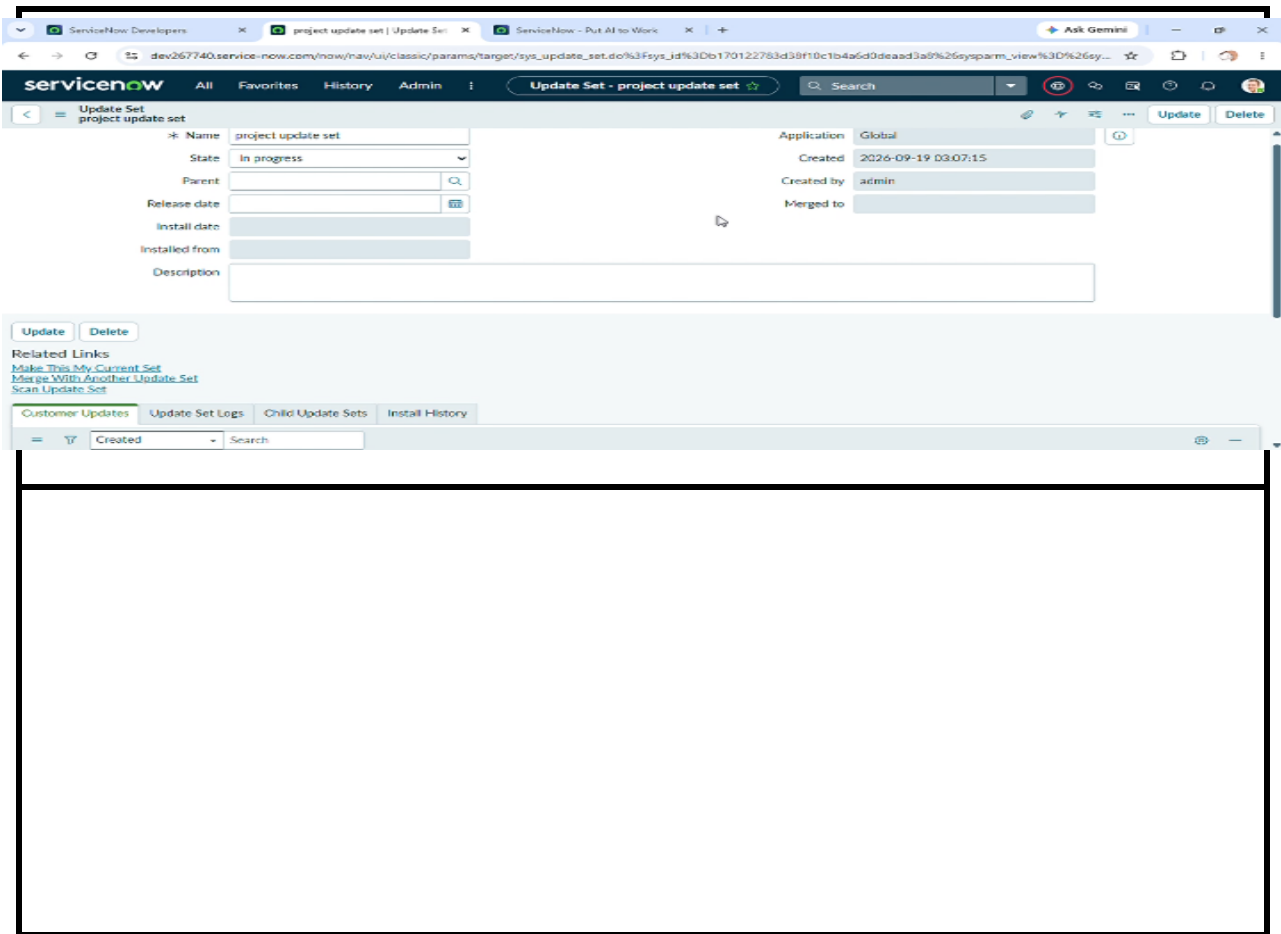
The following table represents the data shown in the 'Update Sets' list view in the second screenshot.

Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
Default	Global	In progress		2026-06-12 14:20:10	system	(empty)	(empty)

The third screenshot shows the 'New record' form for an 'Update Set' with the following fields and values:

- Name: New Update Set
- State: In progress
- Parent: (empty)
- Release date: (empty)
- Description: (empty)
- Application: Global

Buttons at the bottom of the form include 'Submit' and 'Submit and Make Current'.



Phase :- 2 Backend Development & Configuration

Activity 1 : Creation of Custom Table to Store Ticket Records.

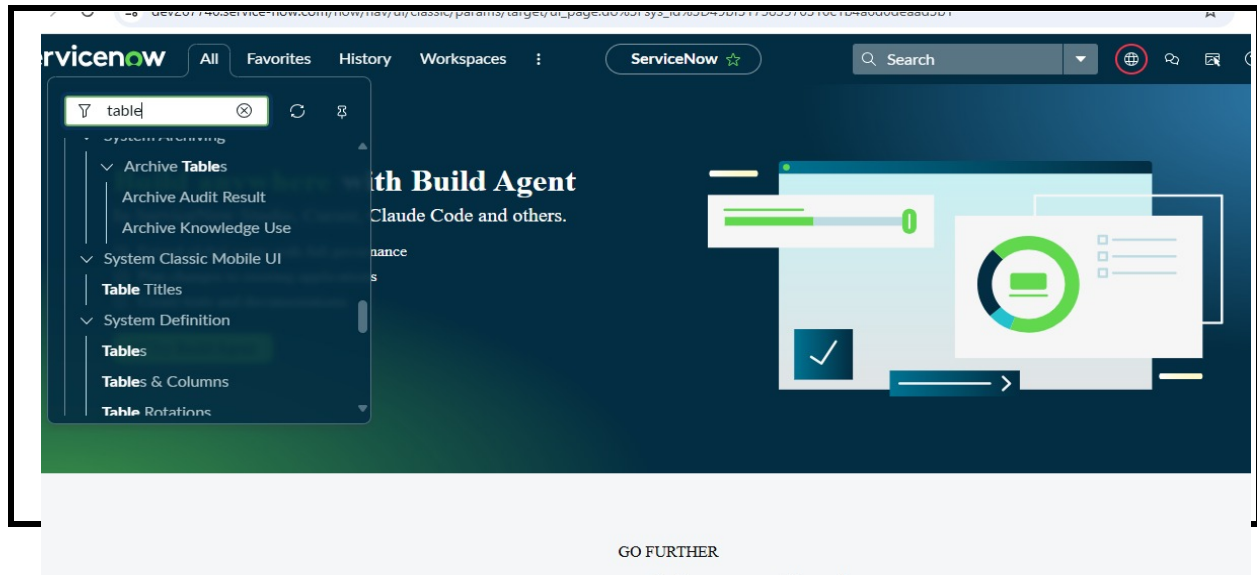
Steps to Create Custom Table

- Open All menus → Search for Tables.
- Under **System Definition** → Click **Tables**.
- On the top right corner Click **New** button.
- Label : **Incident WorkFlow**.
- Uncheck the **Create module**.

Now **Save** the Form.

Then Click the **Controls** Suction and Enable the Check box **true Auto Number**.

Now again **save** the Form.



The screenshot shows the 'Tables' list view in ServiceNow. The table has columns: Label, Name, Extends table, Extensible, and Updated. The table lists various system tables, with 'AIG Access Policy Scope Mapping' selected. The 'Updated' column shows dates from 2026-06-12 14:03:39 to 2026-06-12 14:52:23.

Label	Name	Extends table	Extensible	Updated
Account Subscription Entitlement	account_subscription_entitlement	empty!	false	2026-06-12 14:52:23
Adaptive Authentication Event	adaptive_auth_event	empty!	false	2026-06-12 14:41:06
Agent Assist Recommendation	agent_assist_recommendation	Application File	false	2026-06-12 14:35:51
MLD Server File	agent_file	empty!	false	2026-06-12 14:22:34
AIG Access Policy	aig_access_policy	Application File	false	2026-06-12 14:01:27
AIG Access Policy Scope Mapping	aig_access_policy_scope_mapping	Application File	false	2026-06-12 14:01:28
AIG Scope Tool Mapping	aig_scope_tool_mapping	Application File	false	2026-06-12 14:01:26
Record Producer Configuration	aisa_rp_config	Application File	false	2026-06-12 14:39:11
Search Actions	aisa_ui_action	Application File	false	2026-06-12 14:39:11
AI Search ACL Overrides	ais_acl_overrides	Application File	false	2026-06-12 14:03:37
AI Search Active Table Ingestion Tracker	ais_active_table_ingestion_tracker	empty!	false	2026-06-12 14:03:35
AI Search Transaction Log	ais_api_transaction_log	empty!	false	2026-06-12 14:03:39
AI Search Async Genius Result	ais_async_genius_result	empty!	false	2026-06-12 14:03:39
AI Search Async Request	ais_async_request	empty!	false	2026-06-12 14:03:40

dev267740.service-now.com/now/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_db_object%26syspar...

servicenow All Favorites History Workspaces Table - New Record Search

Table
New record

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Incident WorkFlow

* Name u_incident_workflow

Extends table

Application Global

Create module ☒

Create mobile module ☒

Add module to menu -- Create new --

New menu name Incident WorkFlow

Remote Table ☐

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

Then Save the Form and Under Related Links Click **Show Form**.
Now Open the Configure On clicking the Context Menu and then Open the **Form Designer**.
Then Create Fields as Required.

The screenshot displays the ServiceNow configuration interface for a table named 'Incident Work Flow'. The 'Table - Incident Work Flow' header is visible. Below the header, there are input fields for 'Label' (Incident Work Flow), 'Name' (u_incident_work_flow), 'Application' (Global), and 'Remote Table'. A description box states: 'A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)'. Below this, there are tabs for 'Columns', 'Controls', and 'Application Access'. The 'Controls' tab is selected, showing options for 'Extensible' and 'Live feed'. A blue box contains the text: 'Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format'. Below this, the 'Auto number' section is expanded, showing a 'Prefix' of 'INC' and a 'Number' of '1000'.

Activity 2: Field Creation and Data Type Configuration

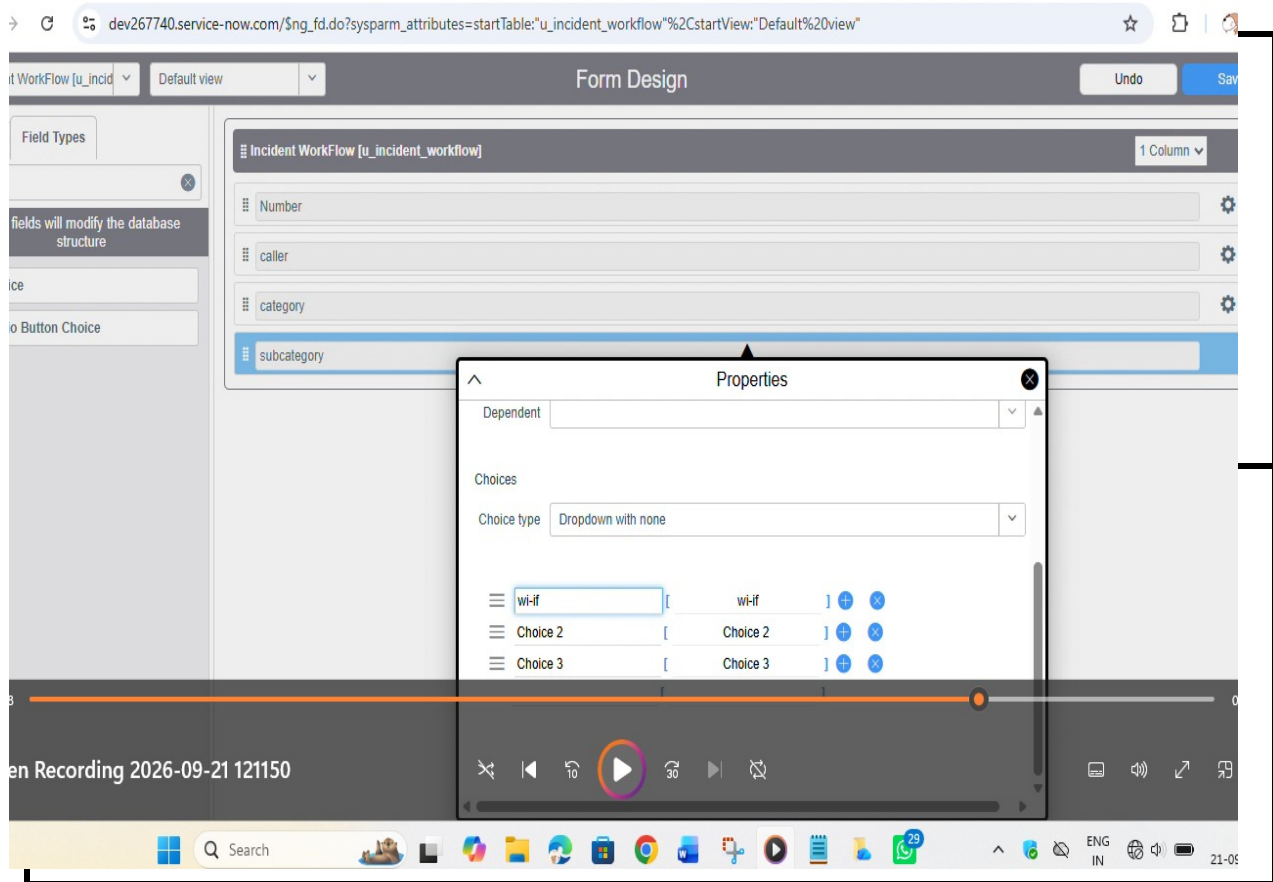
The objective of this milestone is to create all required fields in the School IT Ticket table with correct data types, choices, and dependencies, ensuring structured and consistent data capture.

Field Label	Type	Reference/ Choice
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	[Network,Hardware,Access, Performance]
Subcategory	Choice	[Wi-Fi,Projector,Forgot Password,Slow Computer]
Short Description	String	—
Description	String	—
State	Choice	[New,In progress,On hold,Resolved,Closed]
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

After Creating Fields with their appropriate Type, then add the Choice from the Form design.

To all the Choice list Fields.

They are Category, Subcategory and State.



Same like that, add Choice to the State field.
Then Click the save button on the top right corner.
Open the Form and reload it.

Activity 3 : Implementing Dependency Between Category and Subcategory Choice Fields

The objective of this milestone is to implement a **dependency relationship** between the **Category** and **Subcategory** fields so that the available Subcategory options dynamically change based on the selected Category.

Navigation Steps :

Open the Created Table Form and Right click on the **Subcategory** Field.

Click on the **Configure Dictionary**.

Under **Related links** Click on the **Advanced view**,

Then select the **DependentField** Section.

Enable the checkbox **Use dependent field true**.

Then select the **Field Category** on the **Dependent on field**.

Then **save** the Form.

Dictionary Entry - subcategory View: Advanced*

Related Links
[Show Table](#)
[Run Point Scan](#)

Access Controls Choices (4) Attributes (1) Labels (1)

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
wifi	wifi	en	0	false	2026-09-20 08:06:50	network
slow computer	slow computer	en	3	false	2026-09-20 08:07:33	performance
projector	projector	en	1	false	2026-09-20 08:07:48	hardware
forgot password	forgot password	en	2	false	2026-09-20 08:06:37	access

1 to 4 of 4

Now scroll down to the choice section.

Then add the dependent fields for which option dependent on the category gives the category choice backend name to the subcategory choice values.

Wi-Fi → Network.

Projector → Hardware.

Forgot password → Access.

Slow computer → Performance.

dev267740.service-now.com/u_incident_work_flow.do?sys_id=a57eabcc836f8350c1b4a6d0deaad3ab

Incident Work Flow
INC0001002

Number INC0001002

category network

caller Abraham Lincoln

subcategory wifi

short description -- None --

description projector

sate forgot password

assigend group slow compter

assigend to

Update Delete

Then save the Form.

Then it will display each choice to each one separately based on the category subcategory displays.

dev267740.service-now.com/u_incident_work_flow.do?sys_id=a57eabcc836f8350c1b4a6d0deaad3ab

Incident Work Flow
INC0001002

Number INC0001002

category network

caller Abraham Lincoln

subcategory wifi

short description -- None --

description projector

sate forgot password

assigend group slow compter

assigend to

Update Delete

Incident Work Flow
INC0001002

Number: INC0001002

category: network

caller: Abraham Lincoln

subcategory: wi'fi

short description: -- None --

description: projector

sate: forgot password

assignend group: slow compter

assignend to:

Update Delete

Milestone 3 : Automation using Flow Designer & Email Notif ication.

When a student creates a new IT support ticket, they only need to enter the Caller Name and Short Description of the issue. The student does not need to manually select the Category or Subcategory. Once the ticket is submitted, a Flow Designer automation is

triggered and evaluates the Short Description for predefined keywords such as WiFi, Projector, Password, or Slow. Based on the identified keyword, the flow automatically assigns the appropriate Category and Subcategory to the ticket.

If the short description contains the word *WiFi* , the flow assigns **Category = Network** and **Subcategory = WiFi** .

If the description contains *Projector* , the flow assigns **Category = Hardware** and **Subcategory = Projector** .

If the description includes *Password* or *Login*, the flow assigns **Category = Access** and **Subcategory = Forgot Password** .

If the description mentions *Slow* or *Hanging*, the flow assigns **Category = Performance** and **Subcategory = Slow Computer**.

This automation ensures consistent classification, eliminates human error, and significantly reduces the manual effort required by IT staff .

The entire process is implemented using **Flow Designer** , making the solution easy to maintain, scalable, and completely no-code.

Activity : 1 Auto Classification using Flow design.

Go to **All menus** → Search **Flow designer**.

Under **Process Automation** Open the **Flow designer**.

On the top right corner Click **New** and Select **Flow** Option.

Name : **Auto Classification School IT Tickets**.

Application : **Global**.

Now Click the **Build Flow** button.

ServiceServiceHo xsubcateNew FloForm DprojectAuto CILCOTINC00subcate+Ask Gemini

dev267740.service-now.com/now/workflow-studio/home/flow

Workflow Studio

HomepageOperationsIntegrations

PlaybooksFlowsSubflowsTriggersActionsDecision tables

New

Playbook

Flow

Subflow

Trigger

Action

Decision table

Help me decide

Pick up where you left off

Auto Classify School Tic...
Last updated: 2 h. ago by System ...

Auto classify School Tic...
Last updated: 3 h. ago by System ...

auto classify school ticket
Last updated: 3 h. ago by System ...

Latest updates

System Administrator modified
Auto Classify School Tickets
2 h. ago

System Administrator modified
Auto classify School Ticket
3 h. ago

System Administrator modified
auto classify school ticket
3 h. ago

72

Last refreshed just now.

Name	Application	Status	Active	Updated
Auto Classify School Tickets	Global	Published	true	2026-09-21 13:32
Auto classify School Ticket	Global	Draft	false	2026-09-21 13:32
auto classify school ticket	Global	Draft	false	2026-09-21 13:32
Benchmark Recommendation Evaluator	Benchmarks Spoke	Published	true	2026-06-12 14:42:42
Business process approval flow	Global	Published	true	2020-09-27 22:06:13
Change - Cloud Infrastructure - Authorize	Global	Published	true	2020-11-11 07:08:05
Change - Conflict Detection	Global	Published	true	2026-06-12 14:33:00
Change - Emergency - Authorize	Global	Published	true	2020-10-06 05:39:49

ServiceServiceNe xsubcateNew FloForm DprojectAuto CILCOTINC00subcate+Ask Gemini

dev267740.service-now.com/now/workflow-studio/builder%3FtypeSysId%3D2d85e527439231106c4bb0117fb8f208%26sysId%3D-1%26

Workflow Studio

New Flow
Flow • None

Let's get the details for your flow

Flow name *
Auto classify school Ticket

Application *
Global

Description
Describe your flow.

Show additional properties

Cancel

Build flow

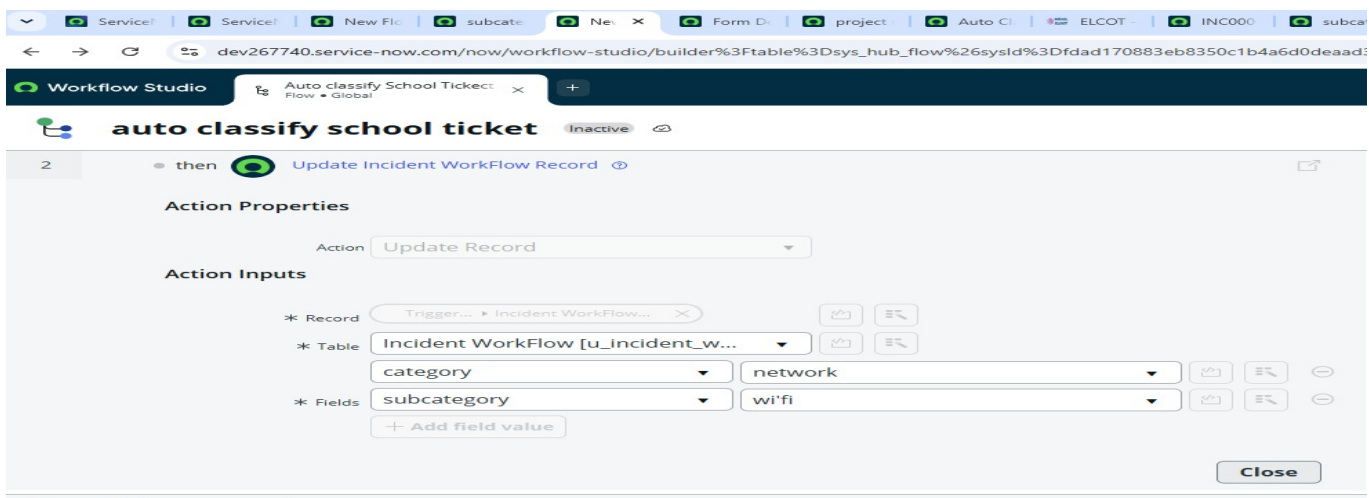
Activity : 2 Trigger Configuration.

Trigger : Record Created.

Table : Incident Workflow.

Condition : Category is Empty.

Then Click Done.



Now Under **Actions** Select **Flow Logic** Option.

Activity : 3 Wi-Fi Issue Classification.

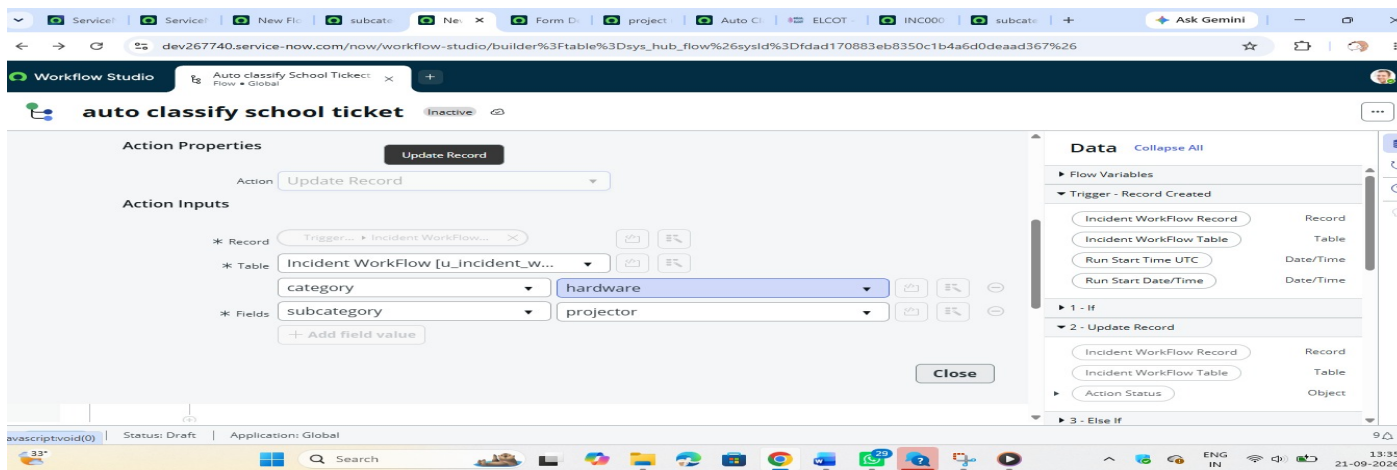
Select **If** condition.

Condition label : SD is **Wifi or Network** (Give as ur wish).

Condition 1 : Using data pill picker Select **Short description** as Field , and operator **Contains** , Value is **Wi-Fi** .

Condition 2 : Using data pill picker Select **Short description** as Field, and Operator is **Contains**, Value is **Network**.

Then Click **Done** Button.



Now Click **Action** to Update record.

Action Inputs.

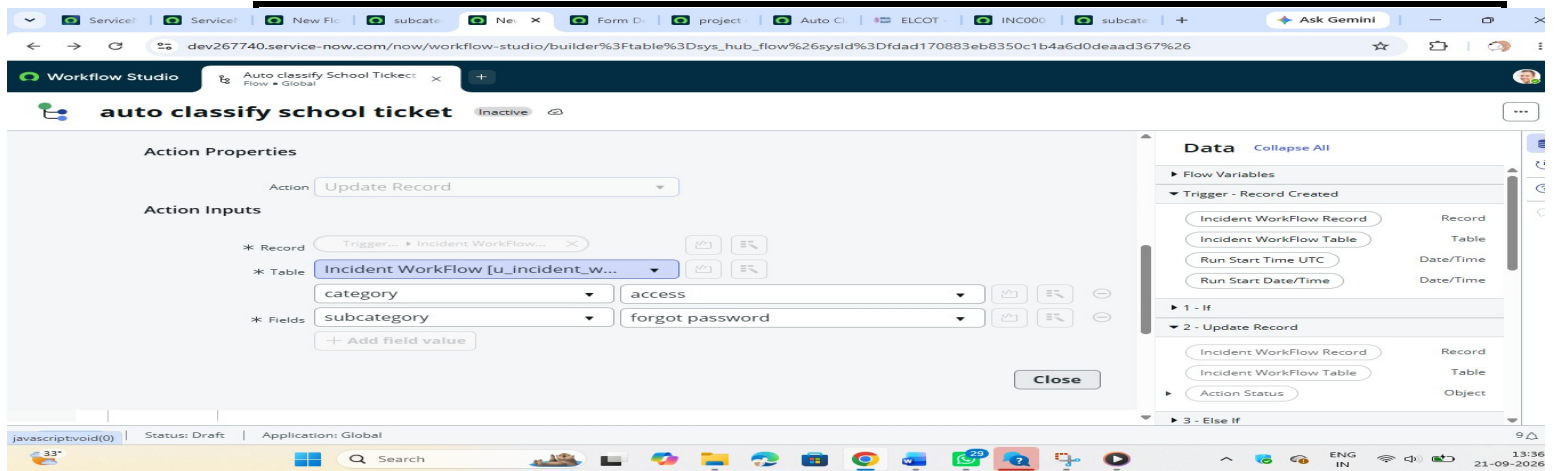
In Record Field drag and drop data from the right side.

Table : Auto updates as Incident Workf low.

Fields Set **Category** = **Network** .

Set **Subcategory** = **Wi-Fi**.

Then Click the **Done** button.



Activity : 4 Projector Issue Classification.

Now Select the **Else If** Condition.

Condition label : SD is **Projector** or **Hardware** .

Condition : Short description Contains **Prjector** .

Then Click the **Done** button.

Now Action is an **Update record**.

Set **Category** = **Hardware** .

Set **Subcategory** = **Projector**.

Workflow Studio

Auto classify School Ticket
Flow • Global

auto classify school ticket Inactive

Action Properties

Action: Update Record

Action Inputs

* Record: Trigger... Incident WorkFlow...

* Table: Incident WorkFlow [u_incident_w...]

* Fields: category performance, subcategory slow computer

+ Add field value

Close

Data Collapse All

Flow Variables

Trigger - Record Created

Incident WorkFlow Record Record

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

Read-only | Status: Draft | Application: Global

33° Search 29 ENG IN 13:36 21-09-2026

Activity : 5 Password Issue Classification.

Now again select **Else If** condition.

Condition label : SD is **Forgot password**.

Condition : **Short description contains Forgot password**.

Then Click the **Done** button.

Now Action is an **Update Record**.

Set **Category** = **Access** .

Set **Subcategory** = **Forgot Password**.

Then Click the **Done** button.

dev267740.service-now.com/now/workflow-studio/builder%3Ftable%3Dsys_hub_flow%26sysId%3D6150a40883274350c1b4a6d0deead3c2%26

Workflow Studio | New Flow | Flow • None | Auto Classify School Tickets | Flow • Global | Auto Classify School Tickets | Flow execution • None

Auto Classify School Tickets Active

View: [Icon] [Icon] [Icon]

Test Deactivate Activate

If SD is wifi or network

then Update Incident Work Flow Record

Else If SD is projector or hardware

then Update Incident Work Flow Record

Else If SD is forgot password

then Update Incident Work Flow Record

Else If SD is performance

then Update Incident Work Flow Record

Published | Application: Global

29

FNG

13:51

Activity : 6 Slow Computer Classification.

Now Again Click the **Else If** Condition.

Condition label : SD is **Performance**.

Condition : **Short description contains Slow Computer**.

Then Click the **Done** button.

Now Action is an **Update Record**.

Set **Category** = **Performance** .

Set **Subcategory** = **Slow computer**.

Then Click the **Done** button.

The screenshot shows the ServiceNow Workflow Studio interface for a workflow named "Auto Classify School Tickets". The workflow is currently "Active". The main canvas displays a series of steps: a start node, followed by a decision node "If SD is wifi or network", then an action "Update Incident Work Flow Record", and then another decision node "Else If SD is projector or hardware". This pattern repeats for "SD is forgot password" and "SD is performance". Each decision node has a "then" branch leading to the "Update Incident Work Flow Record" action. On the right side, there is a "Data" panel with a list of variables: "Flow Variables", "Trigger - Record Created", and a list of steps numbered 1 through 9, including "1 - If", "2 - Update Record", "3 - Else If", "4 - Update Record", "5 - Else If", "6 - Update Record", "7 - Else If", "8 - Update Record", and "9 - Send Email". The bottom status bar indicates the workflow is "Published" and "Application: Global".

Activity : 7 Email Action Setup.

After classification, an email notification is sent using Flow Designer.

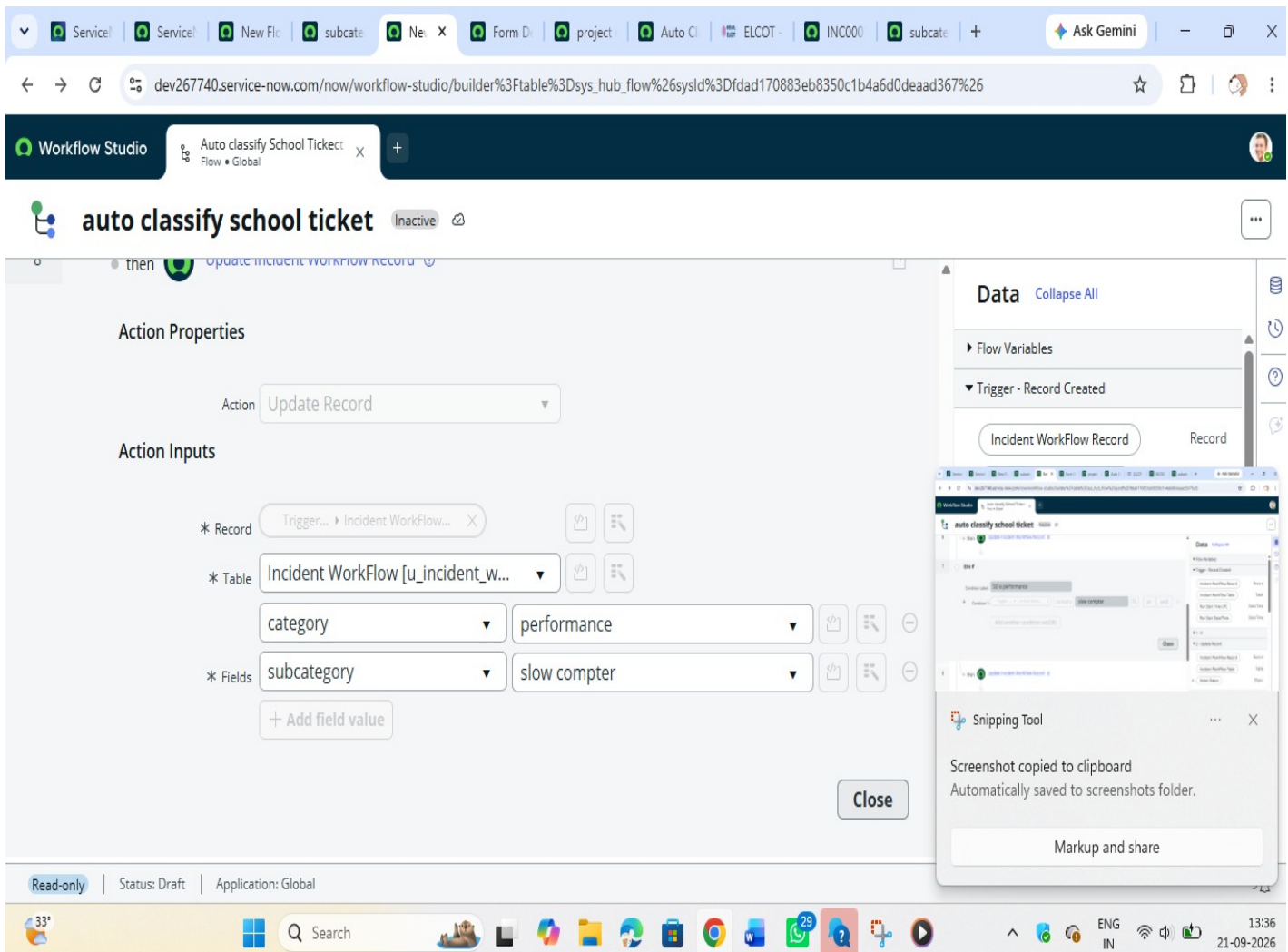
Set Action **Send Email**.

To : **Caller** → **Email**

Subject : **Your Request for the issue has been submitted.**

Body : **Ticket confirmation message.**

Then Click the **Done** button.



Now Click the **Activate** button from the top right corner.

PHASE 6 : Testing, Validation & Security.

The objective of Phase 4 is to ensure that the developed solution works as expected, meets all functional requirements, and follows basic security and access control best practices.

This phase validates the end-to-end automation, confirms data accuracy, and ensures that only authorized users can access or modify ticket data.

Data Validation Checks

Mandatory fields captured correctly.

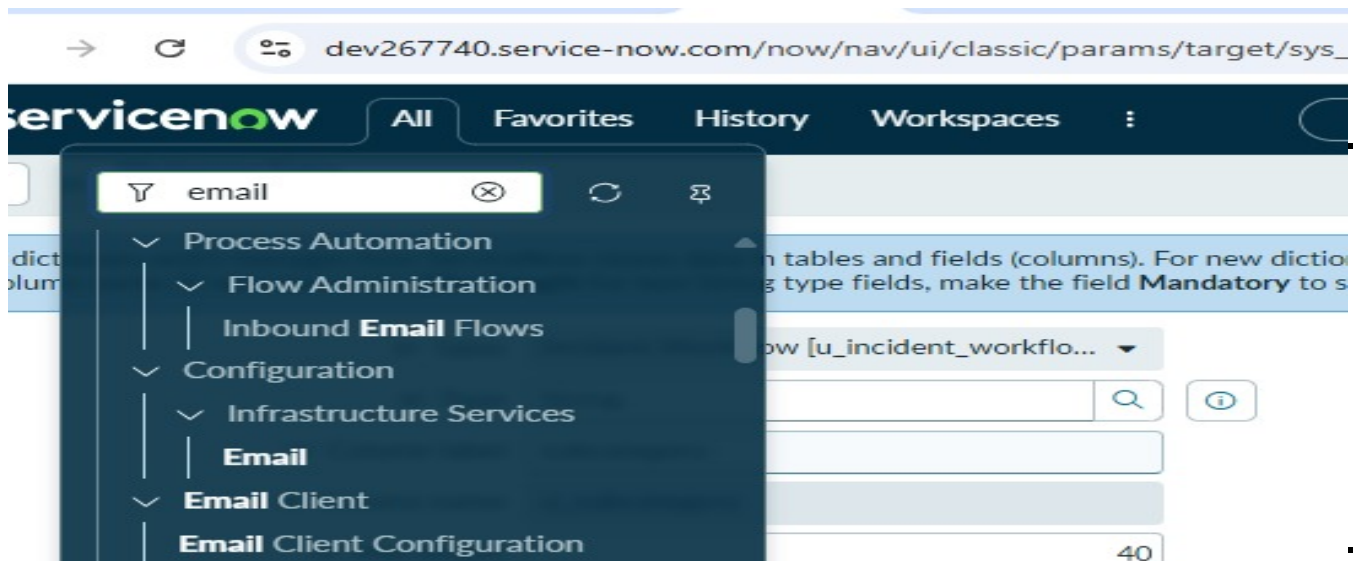
Auto-number generated without duplication.

Category and Subcategory values stored accurately.
Reference fields resolve correctly to user and group tables.

Activity : 1 Testing & Validation.

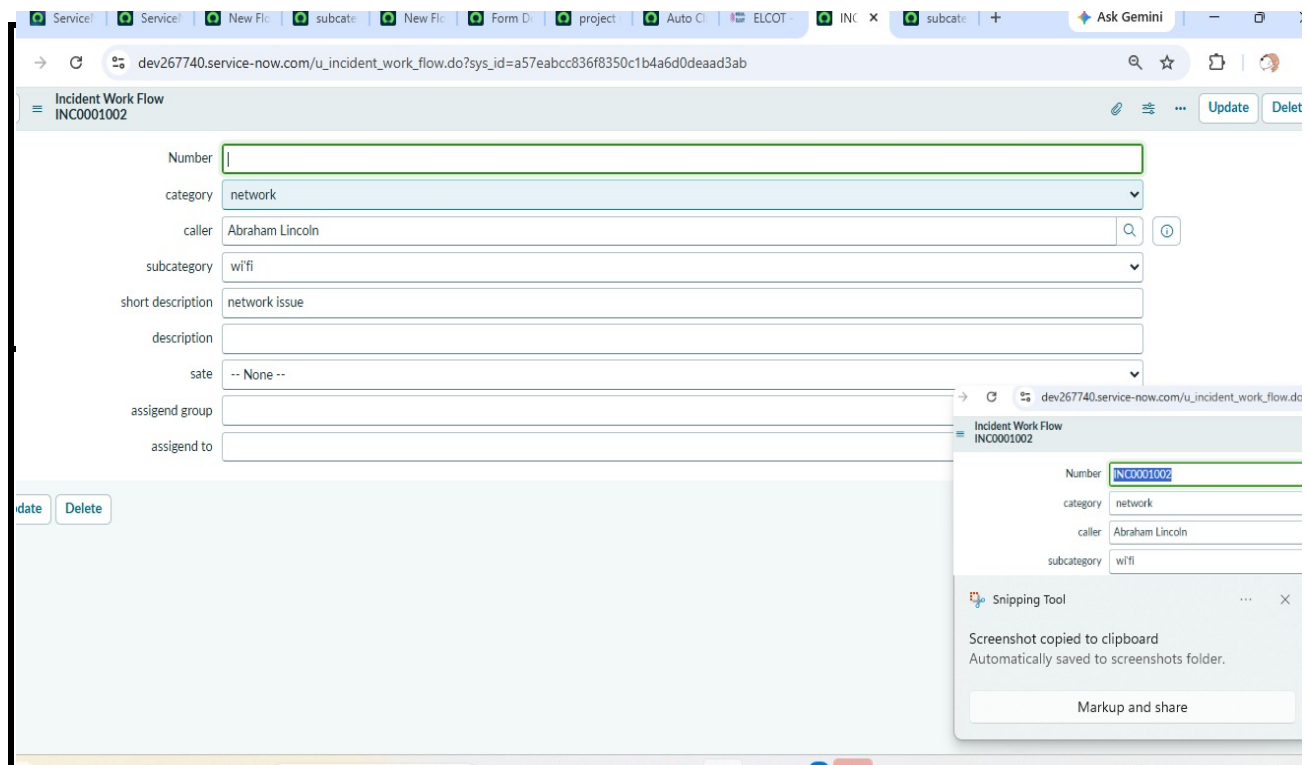
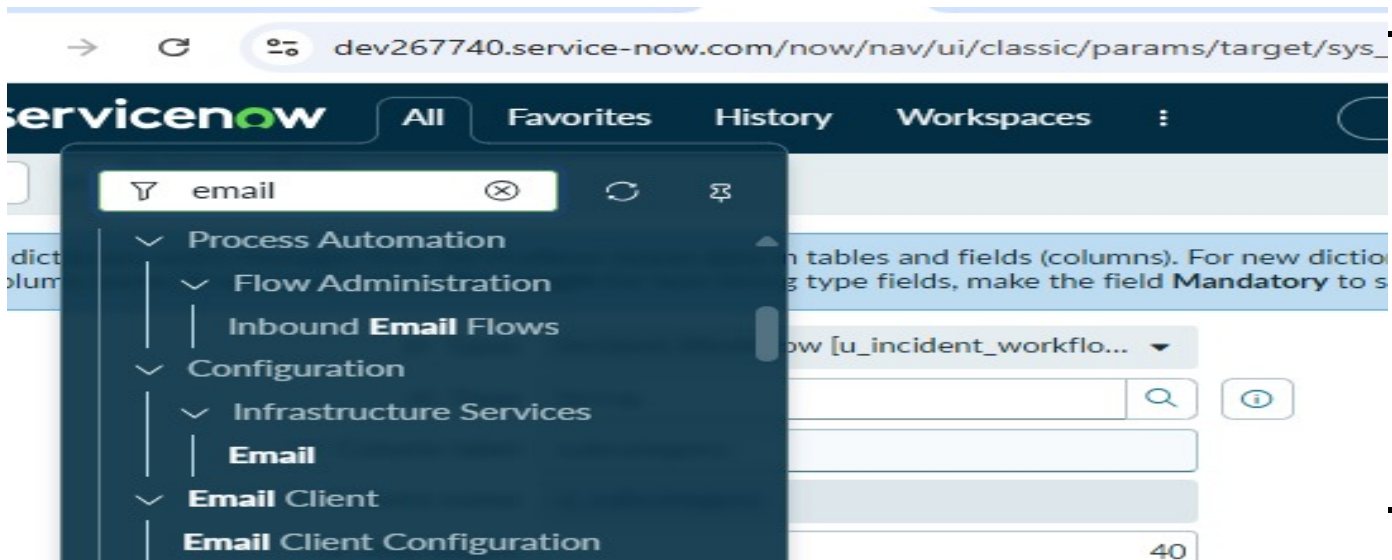
Test Scenario 1

Enter one Caller Name.
Short Description: WiFi not working in library
Then Save the Form.
And Reload the Form.
Then it fills auto as :



Activity : 2 Check Email Notification.

Open All menus → Search Emails.
Under System logs → Select Emails.
Search with the Given Subject Name.
Open that Record and Under Related Links Click Preview mail Option.

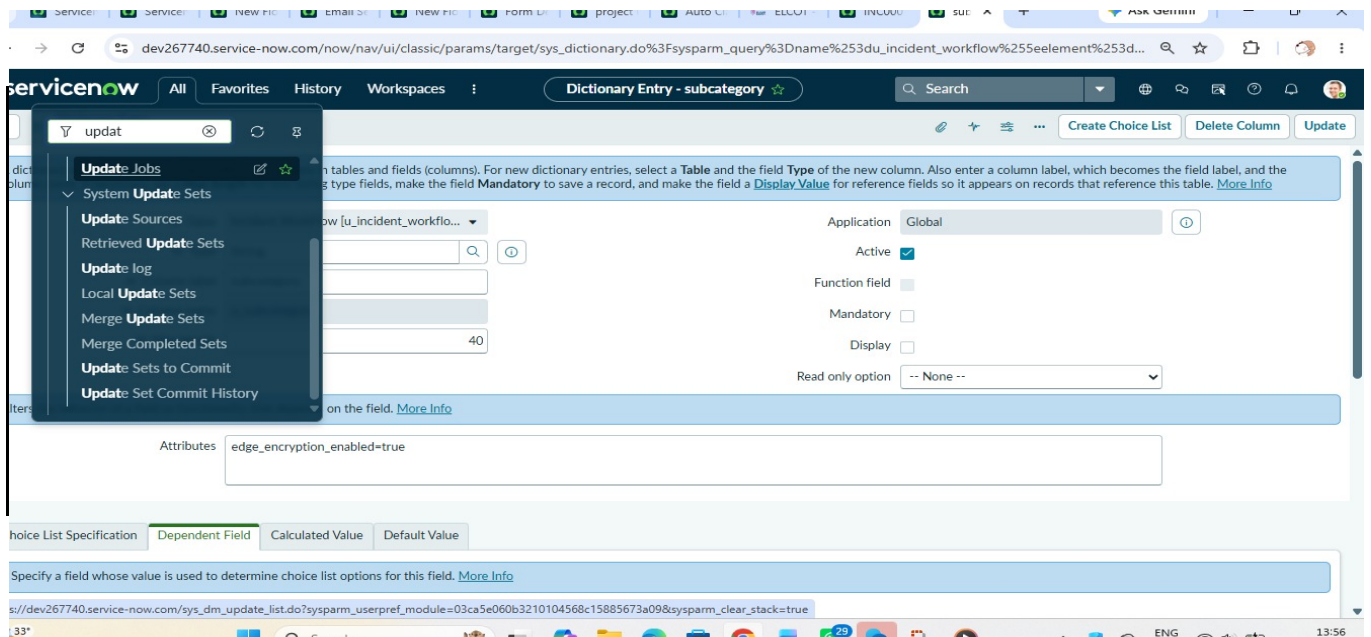


Activity : 3 Test Scenario 2

Enter one Caller Name.

Short Description: Projector not turning on.

Then Save the Form.
And Reload the Form.
Then it fills auto as :
Category → **Hardware**.
Subcategory → **Projector**.
Then Email sent to caller Mail id.



PHASE 5: Deployment & Final Documentation.

The objective of Phase 4 is to deploy the developed solution to the target environment and finalize all project documentation.
This phase ensures that the solution is stable, reusable, and ready for demonstration or production use, along with complete and clear documentation for future reference.

Activity : 1 Make Update set to Complete State.

Go to All menus → Search **Local Update Sets**.
Under **System Update sets** Select → **Local Update sets**.
Open the **Project Update set** .

Make the State From **In progress** to **Complete**.

Then **Save** the Form.

Under Related links Click **Export to XML** Option.

Then the Xml Format the Update set data will be **downloaded** to your system.

We can share through WhatsApp or Mail that downloaded file.

Then the Created Update set will transform to others in this way.

The screenshot shows the ServiceNow interface for a 'Dictionary Entry - subcategory'. The 'Update' dropdown menu is open, displaying a list of update-related options. The main form contains several fields: 'Application' (Global), 'Active' (checked), 'Function field', 'Mandatory', 'Display', and 'Read only option' (None). The 'Attributes' field contains the text 'edge_encryption_enabled=true'. The 'Choice List Specification' tab is active, showing 'Dependent Field' as the selected option. The bottom of the screen shows the browser address bar with the URL 'dev267740.service-now.com/sys_dm_update_list.do?sysparm_userpref_module=03ca5e060b3210104568c15885673a098&sysparm_clear_stack=true'.

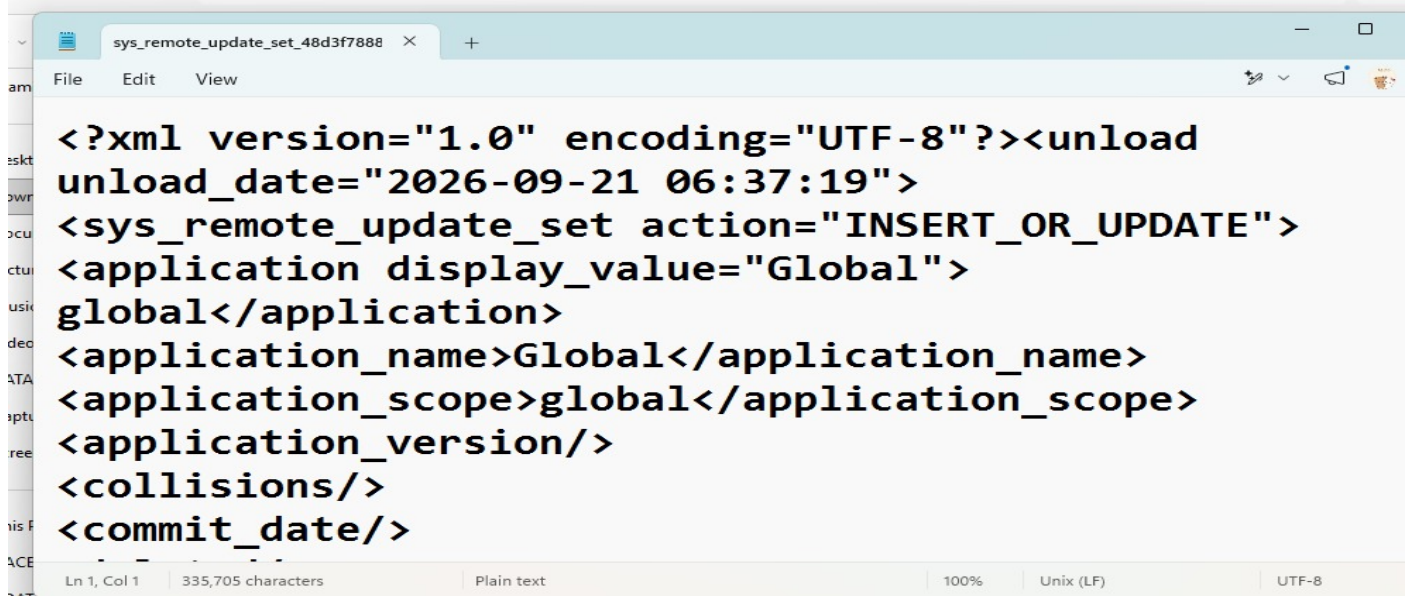
* Name	project update set	Application	Global
State	Complete	Created	2026-09-19 03:07:15
Parent		Created by	admin
Release date		Merged to	
Install date			
Installed from			
Description			

date

Back Out

ited Links

- [rt to XML](#)
- [ge With Another Update Set](#)
- [Update Set](#)



```
<?xml version="1.0" encoding="UTF-8"?><unload
unload_date="2026-09-21 06:37:19">
<sys_remote_update_set action="INSERT_OR_UPDATE">
<application display_value="Global">
global</application>
<application_name>Global</application_name>
<application_scope>global</application_scope>
<application_version/>
<collisions/>
<commit_date/>
```

Conclusion

The **Auto Ticket Classification using Flow Designer** project successfully delivers a complete, end-to-end automation solution for a school IT helpdesk. The primary objective of reducing manual effort in ticket categorization and improving response efficiency has been fully achieved.

Through structured requirement analysis, careful backend configuration, and no-code automation using Flow Designer, the system now automatically classifies IT tickets based on the short description provided by students. The implementation of dependent choice fields ensures data accuracy, while automated email notifications improve communication by instantly acknowledging ticket creation to the caller. The solution was developed using a phased approach, ensuring clarity, stability, and ease of maintenance. Comprehensive testing and security validation confirmed that the system functions reliably, maintains data integrity, and enforces appropriate access controls.

Overall, this project demonstrates how ServiceNow can be effectively used to build scalable, maintainable, and user-friendly automation solutions without relying on complex scripting or machine learning. The design is well-suited for small to medium environments such as educational institutions and can be easily extended in the future to include advanced features like assignment automation, SLA tracking, or predictive intelligence.